

LQR Control of an Inverted Pendulum on a Cart

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EMCH 792: Learning-Based Controls

We reuse the nonlinear Simulink model from Project 1 and add a horizontal cart force. Since designing a controller directly from nonlinear equations is complex, we linearize them near upright. First, we use $\sin(\theta) \approx \theta$ and $\cos(\theta) \approx 1$. Second, we discard the product $\theta\dot{\theta}^2$, which does not contribute to the first-order model. This gives $\dot{z} = Az + BF$ for $z = (x, \dot{x}, \theta, \dot{\theta})^T$. Appendix A gives the derivation.

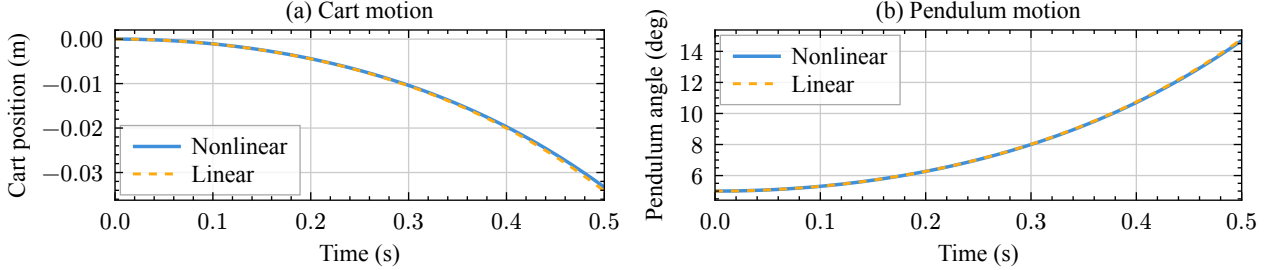


Figure 1: Linear and nonlinear responses to the same unforced 5° release over 0.5 s.

Figure 1 shows that the models respond closely near upright. Their largest differences are 0.094° and 0.87 mm. This supports using the linear model for local controller design. Before designing that controller, we form $C = [B, AB, A^2B, A^3B]$. MATLAB finds rank four, equal to the number of states, so the system is controllable [1].

Since we know the model and have all four states, we choose a linear-quadratic regulator (LQR). A proportional-integral-derivative (PID) controller can instead be tuned from measured response when a reliable model is unavailable [2]. Here, LQR lets us assign a cost to each state error and to the force command [3]. We select weights of 4 for cart position, 2 for cart velocity, 300 for angle, and 10 for angular velocity, collected in $Q = \text{diag}(4, 2, 300, 10)$. The force penalty is $R = 0.5$. These are design choices that emphasize balance while discouraging unnecessary motion and force. Appendix B gives the design and essential MATLAB functions.

MATLAB calculates $u = -Kz$ and confirms that all poles of $A - BK$ have negative real parts. We also simulate the linear closed loop and verify that its states return to zero. We then connect this controller to the nonlinear Simulink model in Appendix E. The actuator limits u to ± 10 N, and a separate disturbance d is added afterward, so the cart receives $F = u + d$.

To test the controller, we run two categories of 30 s simulations. First, we release the pendulum from 5, 10, 20, 30, 45, and 60° to test the linearization and actuator limits. Second, we start upright and apply random cart forces with bounds of ± 2.5 , 5, 7.5, and 10 N. Each force is an independent uniform draw held for 0.1 s. The same seeded sequence is scaled for each bound. Appendix C explains this generation method and its scope. Appendix D gives the full metrics and actuator histories.

RESULTS.

For the initial-angle tests, Figure 2(a,c) shows cart position and pendulum angle over 30 s. Panels (b,d) show 0–6 s details. The 5, 10, and 20° cases recover; the tested 30, 45, and 60° cases fail. The transition from successful recovery to failure to recover therefore lies between the tested 20° and 30° releases.

For the random-force tests, Figure 3(a,c,e) shows cart position, pendulum angle, and applied forces over 30 s. Panels (b,d,f) show 0–3 s details. The controller maintains balance at the tested 2.5, 5, and 7.5 N bounds, but loses balance at 10 N. At 2.5 N, the RMS angle is 0.69° and the peak is 1.79°.

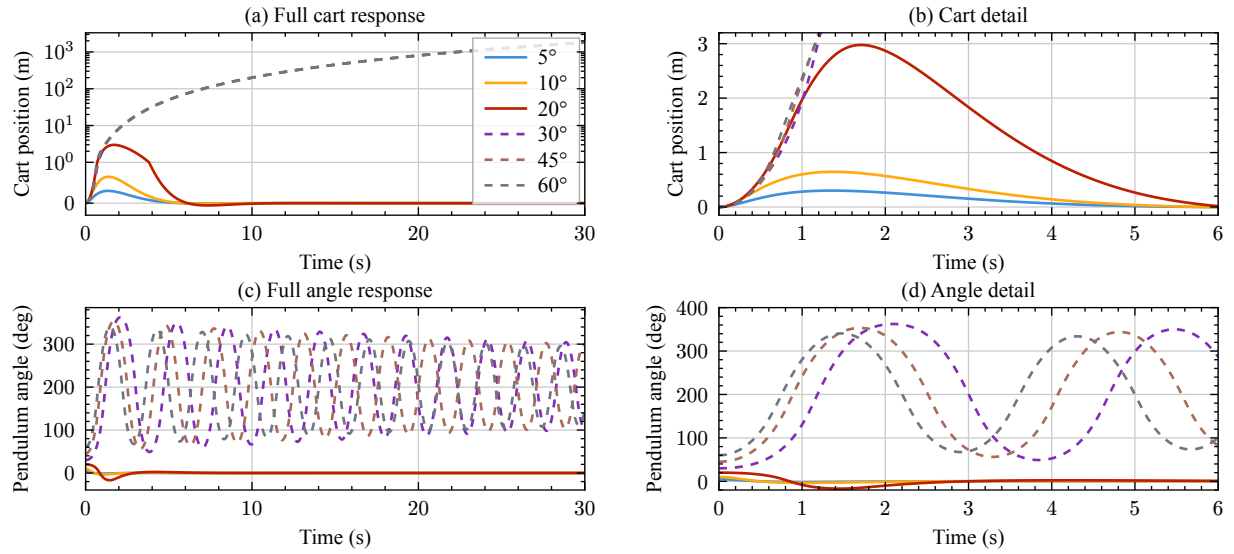


Figure 2: Initial-angle sweep. Dashed traces indicate initial angles where the controller failed to stabilize. Panels (b,d) show 0–6 s. Panel (a) uses a symmetric log scale, linear within ± 1 m.

DISCUSSION.

The smaller tested inputs recover or remain balanced. Failed releases stay at the actuator limit, and the 10 N disturbance also causes sustained saturation. Even successful recovery at 20° needs 2.98 m of cart travel, so a physical track could constrain performance. The sweeps identify an operating range for this gain and force limit. The random-force result applies to the recorded sequence described in Appendix C.

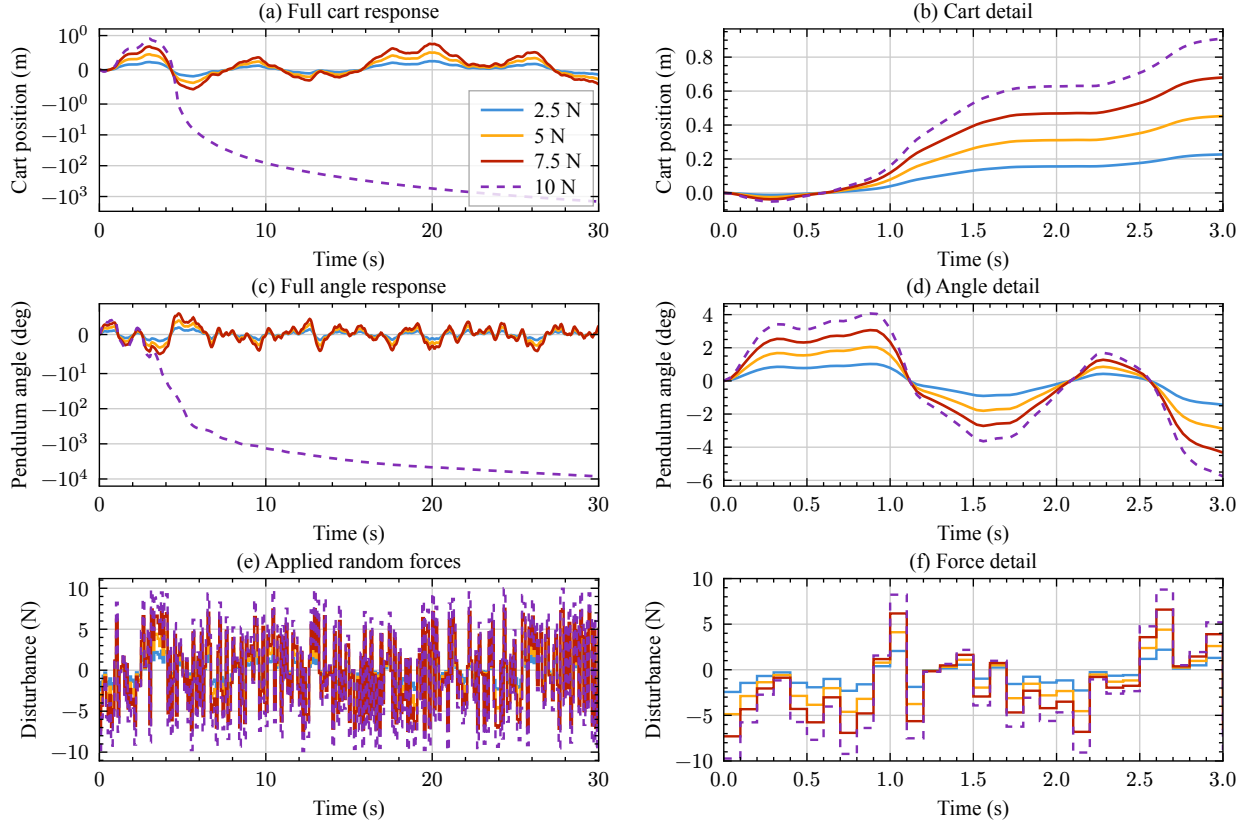


Figure 3: Random-force sweep. Every bound uses the same sequence of force draws; the 10 N case is dashed. Panels (b,d,f) show 0–3 s. Panels (a,c) use symmetric log scales, linear within ± 1 m and $\pm 10^\circ$, respectively.

Appendix A. Nonlinear model, linearization, and model comparisons

We reuse the nonlinear model from Project 1, along with its mechanical assumptions. The cart translates horizontally on a frictionless track. A rigid, massless rod carries a point mass, and a viscous rotational damper acts at the pivot. Motion is planar, gravity is constant, and the track has no end stops. Project 2 adds the horizontal cart force F to the original equations,

$$(M + m)\ddot{x} + m\ell \cos(\theta)\ddot{\theta} - m\ell \sin(\theta)\dot{\theta}^2 = F,$$

$$m\ell \cos(\theta)\ddot{x} + m\ell^2\ddot{\theta} + c_\theta\dot{\theta} - mg\ell \sin(\theta) = 0.$$

Here x is cart displacement, θ is the pendulum angle measured from upright, and dots indicate time derivatives. Positive F acts in the positive x direction. The symbols M , m , ℓ , c_θ , and g denote cart mass, pendulum point mass, rod length, pivot damping coefficient, and gravitational acceleration, respectively. Table 1 gives the values retained from Project 1, using the mass symbols in the equations and MATLAB implementation.

Parameter	Meaning	Value	Unit
M	Cart mass	2.0	kg
m	Pendulum point mass	0.5	kg
ℓ	Rod length	1.0	m
c_θ	Pivot damping	0.01	N m s/rad
g	Gravity	9.81	m/s ²

Table 1: Mechanical parameters reused from Project 1.

Linearization near upright

Because the controller is intended to hold the pendulum upright, we expand about the stationary equilibrium $\theta = \dot{\theta} = \ddot{\theta} = F = 0$. Cart position can be chosen as $x = 0$ because the equations do not depend on absolute position. For angles in radians, the Taylor expansions give

$$\sin(\theta) = \theta - \frac{\theta^3}{6} + \dots, \quad \cos(\theta) = 1 - \frac{\theta^2}{2} + \dots$$

Consequently, $\sin(\theta) \approx \theta$ and $\cos(\theta) \approx 1$ retain the first-order behavior near upright. These approximations become accurate as the angle approaches zero. For example, at 5° , replacing $\sin(\theta)$ by θ introduces approximately 0.13% relative error, and replacing $\cos(\theta)$ by one introduces approximately 0.38% relative error.

We also simplify the centrifugal term, $m\ell \sin(\theta)\dot{\theta}^2$, which represents the horizontal effect of the pendulum's rotational motion on the cart. Near upright, replacing $\sin(\theta)$ by θ makes this term approximately $m\ell\theta\dot{\theta}^2$. It contains three factors that approach zero near the stationary equilibrium. One is the angle θ , and the other two are the angular velocity $\dot{\theta}$ multiplied by itself.

A first-order model retains terms proportional to a single small deviation, such as θ , $\dot{\theta}$, or F . The centrifugal term instead multiplies these deviations together. For example, reducing both the angle and angular velocity by a factor of ten reduces $\theta\dot{\theta}^2$ by a factor of one thousand, while the retained gravity and damping terms decrease by only a factor of ten. Its effect therefore becomes much smaller than the retained terms as the motion approaches upright and rest. We discard it in this local linear model. This step assumes both a small angle and a small angular velocity; a small angle alone does not justify discarding it during rapid rotation. With this term removed and the trigonometric approximations substituted, the equations become

$$(M + m)\ddot{x} + m\ell\ddot{\theta} = F,$$

$$m\ell\ddot{x} + m\ell^2\ddot{\theta} + c_\theta\dot{\theta} - mg\ell\theta = 0.$$

From coupled equations to state space

To separate the accelerations, divide the second linearized equation by $m\ell$ and rearrange it,

$$\ell\ddot{\theta} = -\ddot{x} - \frac{c_\theta}{m\ell}\dot{\theta} + g\theta.$$

Substituting this expression into the first equation eliminates $\ddot{\theta}$,

$$(M + m)\ddot{x} + m\left(-\ddot{x} - \frac{c_\theta}{m\ell}\dot{\theta} + g\theta\right) = F,$$

$$M\ddot{x} = F + \frac{c_\theta}{\ell}\dot{\theta} - mg\theta.$$

Dividing by M and substituting the result back into the angular equation gives

$$\ddot{x} = \frac{F}{M} - \frac{mg}{M}\theta + \frac{c_\theta}{M\ell}\dot{\theta},$$

$$\ddot{\theta} = -\frac{F}{M\ell} + \frac{(M + m)g}{M\ell}\theta - \frac{c_{\theta(M+m)}}{Mm\ell^2}\dot{\theta}.$$

These two second-order equations become four first-order equations by defining

$$z = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{pmatrix} = \begin{pmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{pmatrix}, \quad \dot{z} = \begin{pmatrix} z_2 \\ -\frac{mg}{M}z_3 + \frac{c_\theta}{M\ell}z_4 + \frac{F}{M} \\ z_4 \\ \frac{(M+m)g}{M\ell}z_3 - \frac{c_{\theta(M+m)}}{Mm\ell^2}z_4 - \frac{F}{M\ell} \end{pmatrix}.$$

Collecting the state coefficients and force coefficients gives $\dot{z} = Az + BF$. With the parameters in Table 1, the matrices are

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -2.4525 & 0.005 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 12.2625 & -0.025 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0.5 \\ 0 \\ -0.5 \end{pmatrix}.$$

Before designing the controller, MATLAB forms the controllability matrix $\mathcal{C} = [B, AB, A^2B, A^3B]$ and finds $\text{rank}(\mathcal{C}) = 4$. The rank equals the four-state dimension, so the cart force can control all four states in this linear model.

Comparison procedure

Having obtained the linear model, we compare it with the original nonlinear equations under identical initial conditions and inputs. Both models run without feedback for 1 s using MATLAB's ode45 solver, relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and a maximum step of 0.002 s. The nonlinear calculation uses the same acceleration function as the Simulink plant. No actuator limit is applied to these prescribed-force comparisons.

Because upright is unstable without feedback, 10 or 20 s would primarily compare motion far outside the linearization region. We instead tabulate absolute differences at 0.1, 0.25, and 0.5 s and show the complete first second. The errors are $e_{x(t)} = |x_{\text{lin}(t)} - x_{\text{nonlin}(t)}|$ and $e_{\theta(t)} = |\theta_{\text{lin}(t)} - \theta_{\text{nonlin}(t)}|$, reported in metres and degrees. The curves use matching colors for each input, with solid lines for the nonlinear model and dashed lines for the linear model.

Initial-angle comparisons

First, we release both models from 2, 5, 10, 15, 20, 25, and 30°, with zero cart position, zero velocities, and zero applied force. Figure 4 compares the motion, and Table 2 reports the differences at the selected times.

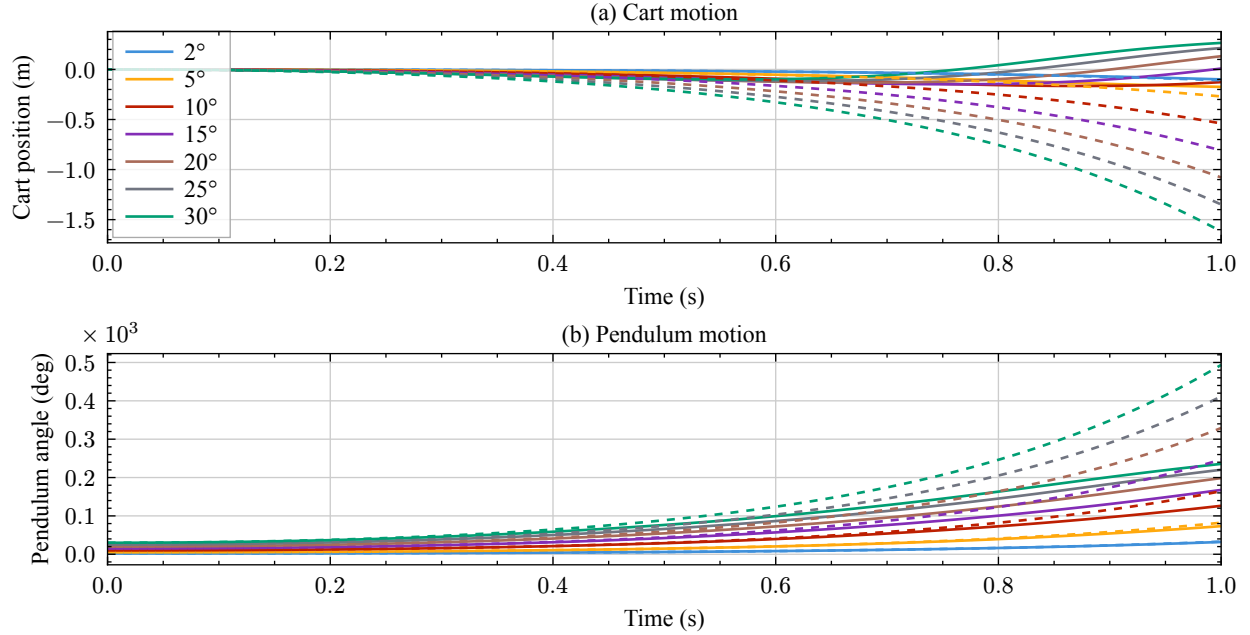


Figure 4: Uncontrolled responses from seven initial angles over 1 s. Solid lines use the nonlinear equations; dashed lines use the upright linearization. Matching colors indicate matching initial angles.

Initial angle (deg)	0.1 s		0.25 s		0.5 s	
	e_x (m)	e_θ (deg)	e_x (m)	e_θ (deg)	e_x (m)	e_θ (deg)
2	5.08×10^{-7}	6.55×10^{-5}	4.43×10^{-6}	0.0005	5.65×10^{-5}	0.0061
5	7.92×10^{-6}	0.001	6.88×10^{-5}	0.0084	0.000866	0.0937
10	6.26×10^{-5}	0.0081	0.000541	0.0664	0.006497	0.7049
15	0.000207	0.0267	0.00177	0.2179	0.019949	2.1728
20	0.000478	0.0618	0.004031	0.4975	0.042204	4.6229
25	0.000902	0.1168	0.007497	0.9284	0.072811	8.0355
30	0.001499	0.1944	0.012247	1.5228	0.110647	12.324

Table 2: Absolute model differences following unforced releases. Each time has cart-position error e_x in metres and angle error e_θ in degrees.

At 0.5 s, the 2° release differs by 0.006° and 0.056 mm. At 30°, those differences increase to 12.32° and 0.111 m. The increasing differences show why the linear model is appropriate near upright and why larger releases require verification with the nonlinear plant.

Applied-force comparisons

Second, we start both models upright and at rest and apply a constant horizontal force from $t = 0$. The six force levels are 0.5, 1, 2, 4, 8, and 16 N. Figure 5 shows the responses, and Table 3 reports the differences. These are prescribed inputs for model validation, separate from the later controller tests with a 10 N actuator limit.

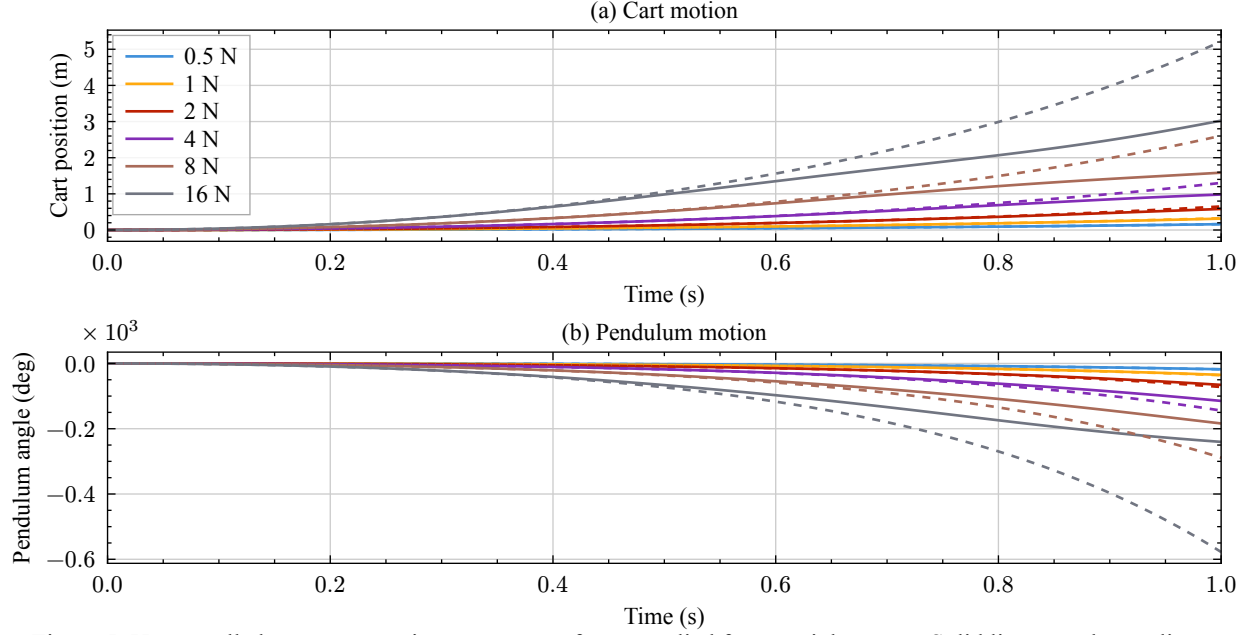


Figure 5: Uncontrolled responses to six constant cart forces applied from upright at rest. Solid lines use the nonlinear equations; dashed lines use the upright linearization. Matching colors indicate matching forces.

Force (N)	0.1 s		0.25 s		0.5 s	
	e_x (m)	e_θ (deg)	e_x (m)	e_θ (deg)	e_x (m)	e_θ (deg)
0.5	1×10^{-10}	9.59×10^{-9}	2.87×10^{-8}	2.74×10^{-6}	3.18×10^{-6}	0.0003
1	8.04×10^{-10}	7.67×10^{-8}	2.29×10^{-7}	2.19×10^{-5}	2.54×10^{-5}	0.0024
2	6.43×10^{-9}	6.14×10^{-7}	1.83×10^{-6}	0.0002	0.000202	0.0193
4	5.14×10^{-8}	4.91×10^{-6}	1.47×10^{-5}	0.0014	0.00158	0.1507
8	4.11×10^{-7}	3.93×10^{-5}	0.000117	0.0112	0.011629	1.1046
16	3.29×10^{-6}	0.0003	0.000919	0.0877	0.072246	6.8103

Table 3: Absolute model differences under constant cart force. Each time has cart-position error e_x in metres and angle error e_θ in degrees.

At 0.5 s, the 0.5 N input gives an angle difference of 0.0003° , while 16 N gives 6.81° . Stronger forces drive the pendulum farther from upright, increasing the effect of the nonlinear terms. Together, the release and force comparisons support using the linear model for local controller design. They also establish the need to test that controller on the nonlinear model, as done in Appendices C and D.

Appendix B. LQR design and essential MATLAB functions

With the controllable model from Appendix A, we choose a gain that minimizes the accumulated state error and force cost,

$$J = \int_0^\infty (4x^2 + 2\dot{x}^2 + 300\theta^2 + 10\dot{\theta}^2 + 0.5u^2) dt.$$

Here x is in metres, θ is in radians, and u is in newtons. The state-error matrix is $Q = \text{diag}(4, 2, 300, 10)$, and the input penalty is $R = 0.5$. These weights were selected to prioritize upright balance, retain a penalty on cart displacement, damp velocity, and discourage excessive force. They are tuning choices, rather than coefficients derived from the mechanical equations.

MATLAB solves the algebraic Riccati equation through its LQR routine. It returns

$$u = -Kz, \quad K = (-2.8284, -6.3204, -86.8641, -25.2346).$$

The poles of $A - BK$ are $-0.699 \pm 0.548i$ and $-4.042 \pm 1.125i$. Their negative real parts establish stability of the linear closed loop. To verify the corresponding time response, we also simulate it from a 5° release with no disturbance. Its final state norm after 30 s is approximately 6.03×10^{-10} .

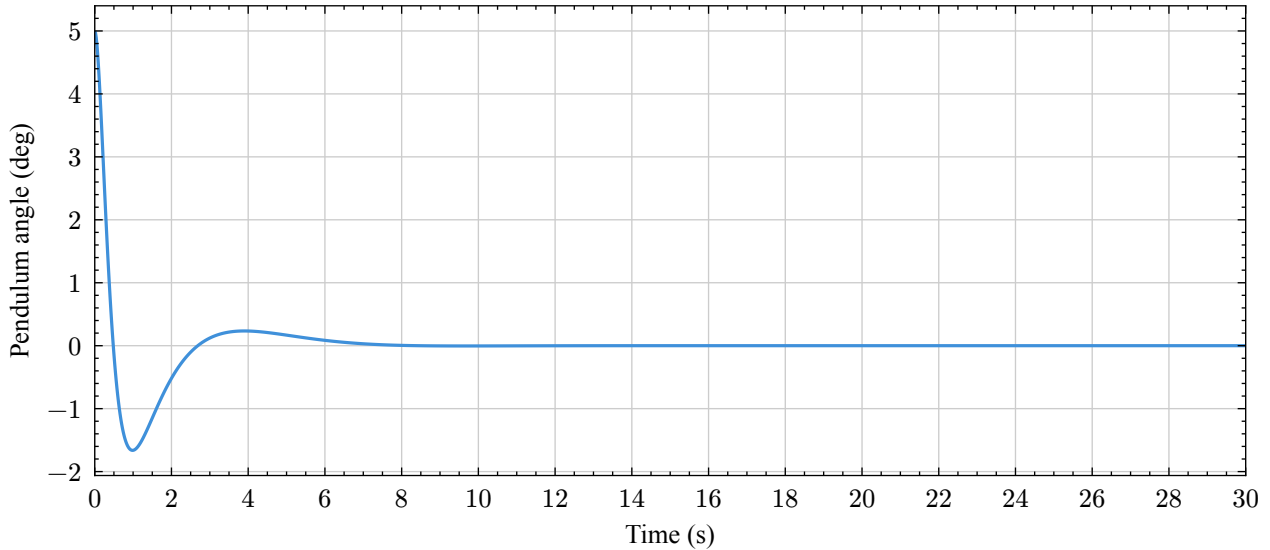


Figure 6: Linear closed-loop response used to verify the designed gain.

The nonlinear model in Appendix E uses this same gain, with its command limited to ± 10 N. The nonlinear sweeps therefore evaluate how the linear design performs when the exact equations and force limit are restored.

Controller and acceleration functions

The preceding design is implemented by the following function. It constructs the matrices, checks controllability, computes the gain, and checks the closed-loop poles.

```
function [K, design] = design_controller(M, m, ell, c_theta, g)
% Design state feedback for z = [x; x_dot; theta; theta_dot].
% Theta is measured in radians from upright; force is in newtons.

A = [0 1 0 0; ...
     0 0 -m*g/M c_theta/(M*ell); ...
     0 0 0 1; ...
     0 0 (M+m)*g/(M*ell) -c_theta*(M+m)/(M*m*ell^2)];
B = [0; 1/M; 0; -1/(M*ell)];

% Selected penalties for position, velocity, angle, angular velocity.
% The input penalty balances state regulation against control effort.
Q = diag([4 2 300 10]);
R = 0.5;
controllability_rank = rank(ctrb(A, B));
assert(controllability_rank == 4, 'The linear model is not controllable.');
```

```
[K, ~, poles] = lqr(A, B, Q, R);
assert(all(real(poles) < 0), 'The linear closed loop is unstable.');
```

```
design = struct('A', A, 'B', B, 'Q', Q, 'R', R, 'K', K, ...
    'poles_real', real(poles), 'poles_imag', imag(poles), ...
    'controllability_rank', controllability_rank);
end
```

The nonlinear Simulink block then uses the function below to calculate the two accelerations. Its eighth input is the total cart force.

```
function [theta_ddot, x_ddot] = accelerations(theta_dot, theta, M, m, ell, c_theta, g, F_cart)
% Nonlinear cart-pendulum equations with an applied horizontal cart force.
% Positive theta is the upright perturbation used in Mini Project 1.

s = sin(theta);
c = cos(theta);
mass_matrix = [M + m, m * ell * c; m * ell * c, m * ell^2];
forcing = [F_cart + m * ell * s * theta_dot^2; ...
    m * g * ell * s - c_theta * theta_dot];
acceleration = mass_matrix \ forcing;

x_ddot = acceleration(1);
theta_ddot = acceleration(2);
end
```

The separate model builder and experiment script remain runnable files alongside the report.

Appendix C. Random-force generation and measurement protocol

The feedback loop in Appendix E receives a separate external force. To generate it, MATLAB draws independent samples ξ_k uniformly in $[-1, 1]$ and holds each value for $\Delta t = 0.1$ s. For force bound a , the applied disturbance is

$$d_{a(t)} = a\xi_k, \quad k\Delta t \leq t < (k+1)\Delta t.$$

We use $a = 2.5, 5, 7.5$, and 10 N. The generator uses the Twister algorithm with seed 79202. Every bound uses the same ξ_k sequence, so the tests change force strength while retaining the timing and sign of each push. A sample beyond 30 s prevents the source block from needing to extrapolate at the final solver step. Interpolation is disabled, and the source holds its last value after the final sample.

This is one model of short, irregular pushes. Its theoretical mean is zero and its variance is $\frac{a^2}{3}$, although the finite sequence has its own sample mean. It does not model a persistent bias, correlated low-frequency force, isolated impact, or sensor noise. A different seed would produce a different trajectory. The results therefore compare the four strengths for this recorded sequence; they do not estimate a failure probability across random realizations.

Both test categories run for 30 s using ode45 with a 0.01 s maximum step. Release tests set the initial angle to 5, 10, 20, 30, 45, or 60°, with all other initial states and the disturbance zero. Random-force tests start with every state zero. The gain and actuator limit stay fixed across all trials.

To compare the trials, the logged states are sampled on a uniform 0.02 s grid. Peak angle is $\max_i |\theta_i|$, and root-mean-square (RMS) angle is

$$\theta_{\text{RMS}} = \sqrt{\left(\frac{1}{N}\right) \sum_i \theta_i^2}.$$

Settling time is the first sampled time after which $|\theta|$ stays within 1° through the end of the run. This describes release recovery; under continuous forcing it depends on the particular later draws. Saturation fraction counts analysis samples with $|u| \geq 10$ N. For release tests, recovery requires the final angle magnitude and the RMS angle over the final two seconds to be below 2°. For random-force tests, peak and RMS angle measure the continuing motion, and loss of upright balance is identified by leaving $\pm 90^\circ$. Figures use every second analysis sample, while metrics use every sample. The forcing panels use the original 0.1 s values and display the actual held steps.

Appendix D. Detailed release and disturbance results

The release tests defined in Appendix C produce the following metrics. The final-two-second RMS checks that a successful final angle is accompanied by sustained balance. The saturation percentage is measured on the analysis grid.

Initial angle (deg)	Peak cart displacement (m)	Final 2 s RMS (deg)	Saturation (%)	Recovered
5	0.301	0	0	Yes
10	0.646	0	0.33	Yes
20	2.977	0	3.93	Yes
30	1799.904	227.543	100	No
45	1800.317	203.3561	100	No
60	1800.069	217.6301	100	No

Table 4: Release-test recovery, cart travel, and actuator saturation.

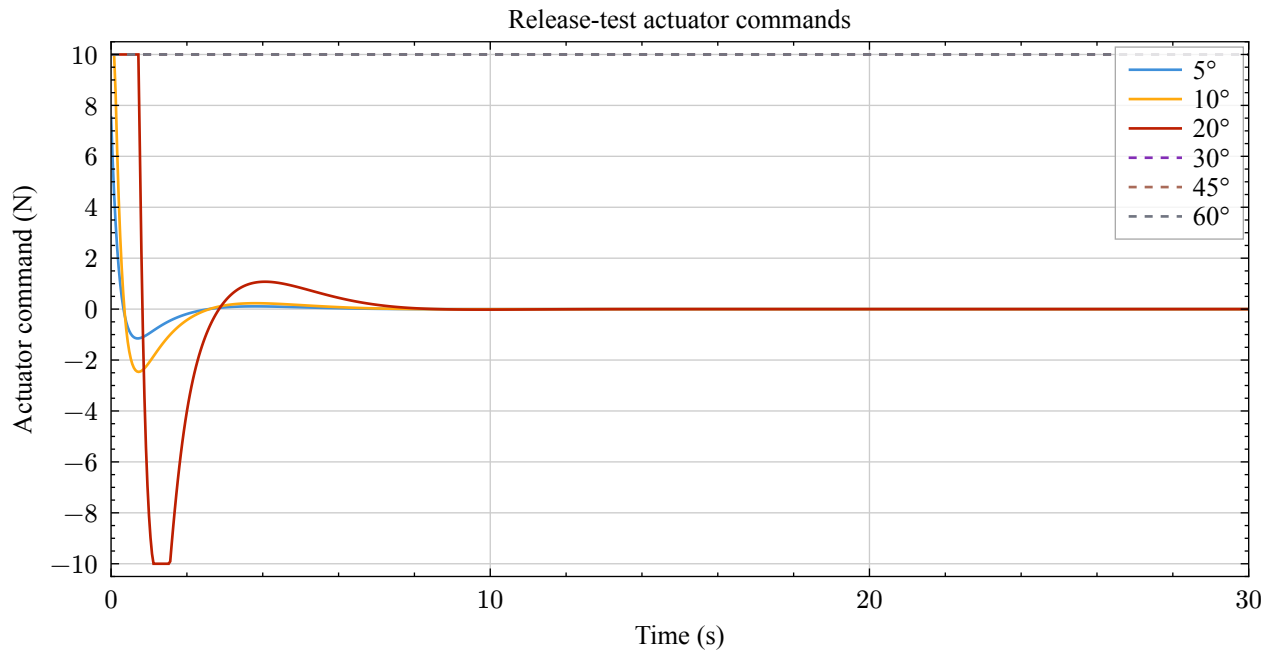


Figure 7: Force used in the initial-angle sweep. The three failed cases remain at the actuator limit.

The tested 20° case recovers but reaches 2.98 m of displacement. The tested 30° case fails, placing the transition somewhere between those two inputs for this controller. No trials between them were run. After failure, the unlimited-track model permits the cart to keep accelerating, and the angle records full rotations rather than being wrapped into a single revolution.

Random-force metrics and actuator use

The force sweep measures continuing balance while the disturbance remains active. Its angle, travel, and actuator metrics are

Bound (N)	RMS angle (deg)	Peak angle (deg)	Peak cart displacement (m)	Saturation (%)	Peak command (N)
2.5	0.69	1.79	0.253	0	3.08
5	1.39	3.59	0.506	0	6.16
7.5	2.08	5.38	0.759	0	9.24
10	4176.16	8429.52	1456.048	88.74	10

Table 5: Random-force balance, cart travel, and actuator demand.

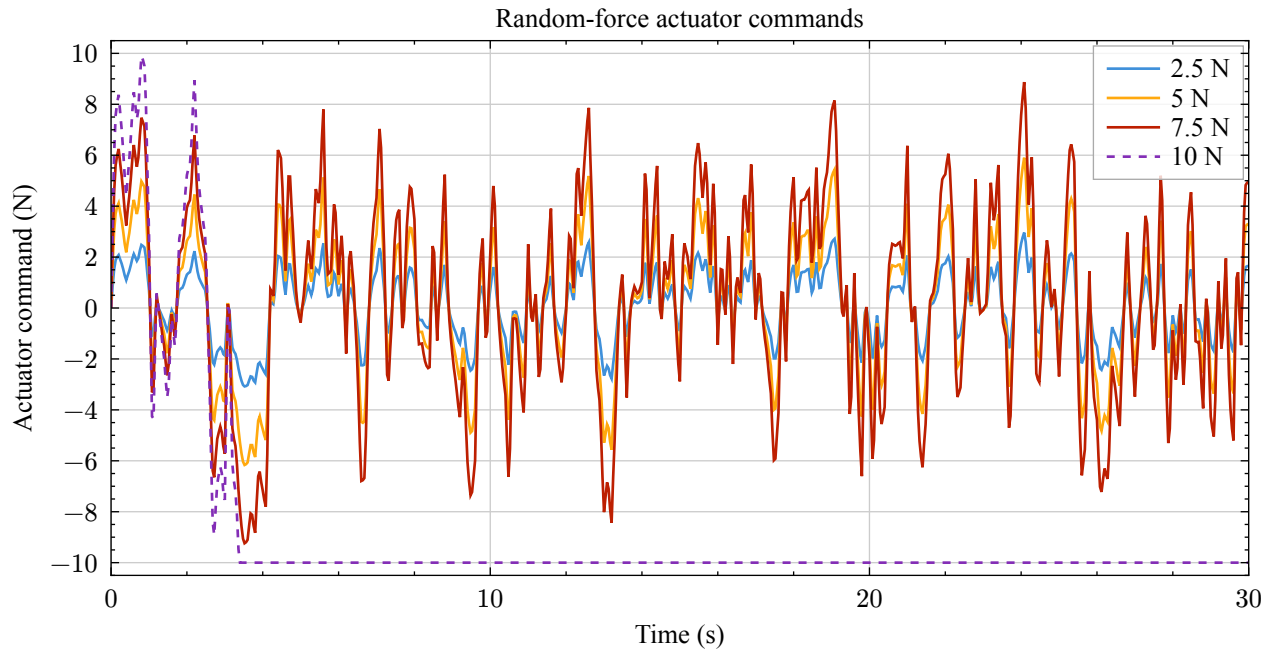


Figure 8: Actuator use under the four disturbance bounds. The dashed 10 N case loses balance and remains saturated after failure.

The held inputs appear over 30 s in Figure 3(e) and over 0–3 s in Figure 3(f). At bounds through 7.5 N, the angle stays within 5.4° and the actuator never saturates. The 7.5 N case ends at 2.29° because the pushes continue; this is ongoing disturbed balance rather than a return to zero. At 10 N, the pendulum leaves the upright region and the command saturates. Once it falls, that limited command cannot restore the local upright response.

Velocity states used by the controller

The preceding figures show position, angle, and force. The remaining two feedback states are cart velocity and angular velocity. The following histories compare the nominal 5° release and 2.5 N random-force test.

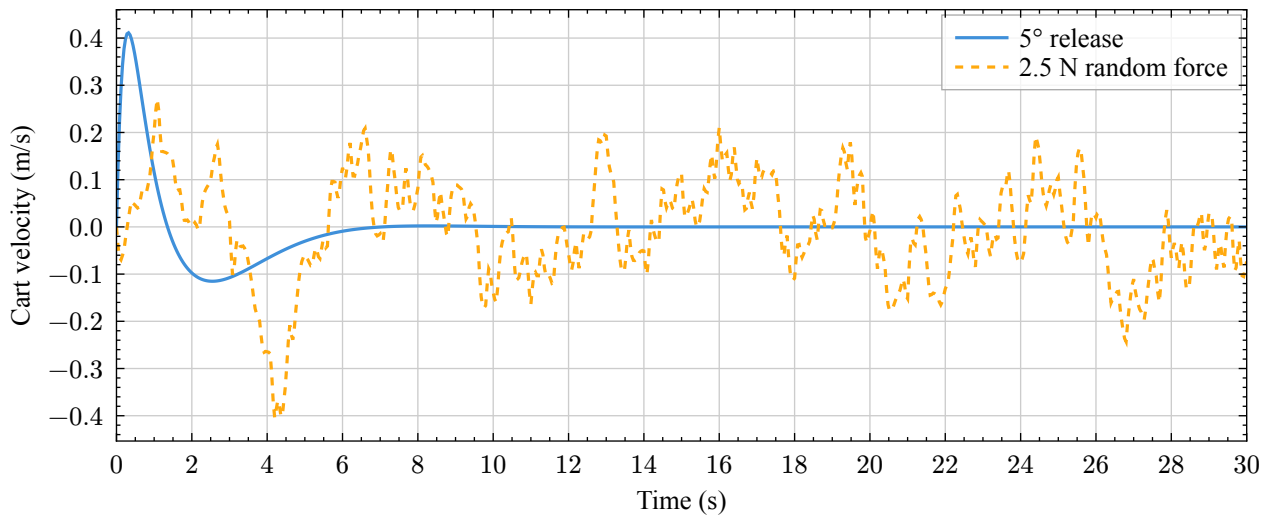


Figure 9: Cart velocity supplied to the state-feedback gain.

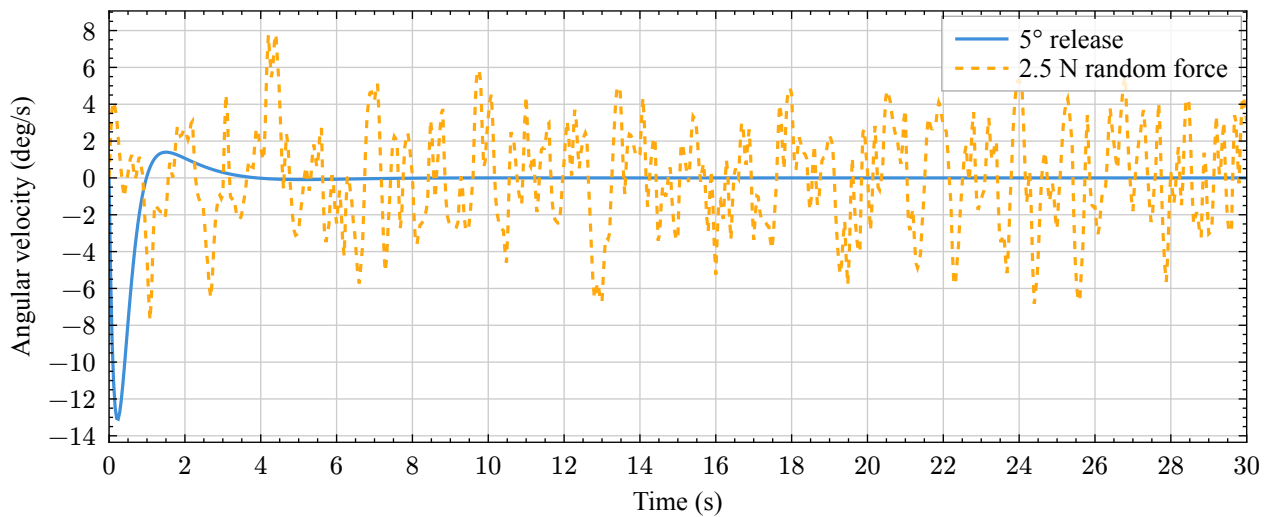


Figure 10: Pendulum angular velocity. The controller uses radians per second internally.

The release velocities decay as the cart and pendulum settle. Under continuing random pushes, the velocities continue to vary while the controller maintains balance. These signals complete the four-state record used to produce the report's position and angle responses.

Appendix E. Simulink feedback loop and nonlinear plant

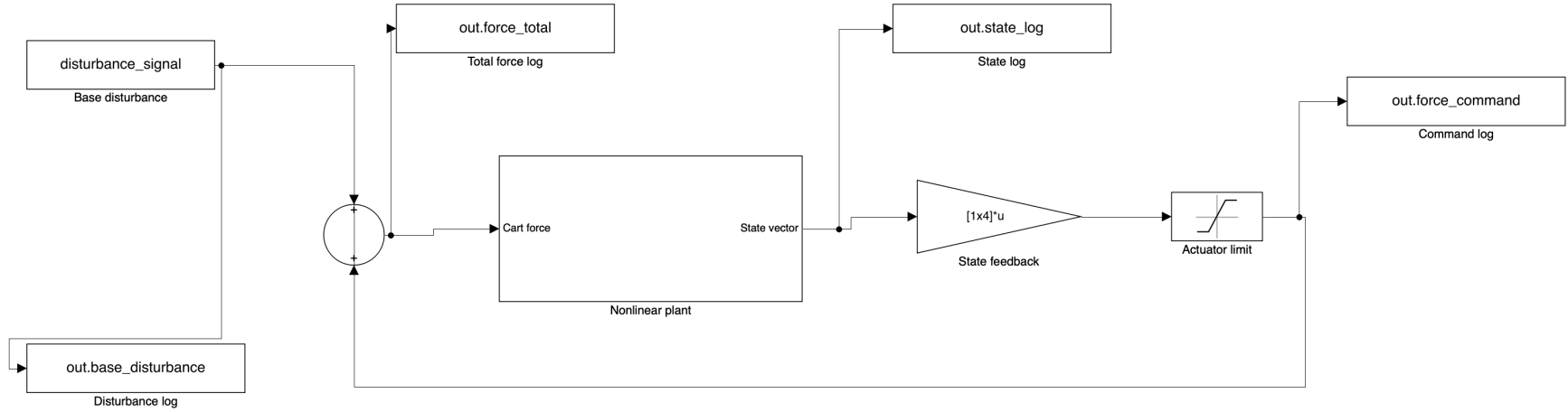


Figure 11: Complete feedback loop. The disturbance enters above the circular sum, and the limited actuator command returns below the loop.

The gain from Appendix B acts on all four simulated states. The actuator block limits the command, then the sum adds the disturbance before the total force reaches the cart. Output blocks record the states, command, disturbance, and total force. This wiring implements $F = \text{sat}(-Kz) + d$.

Nonlinear plant subsystem

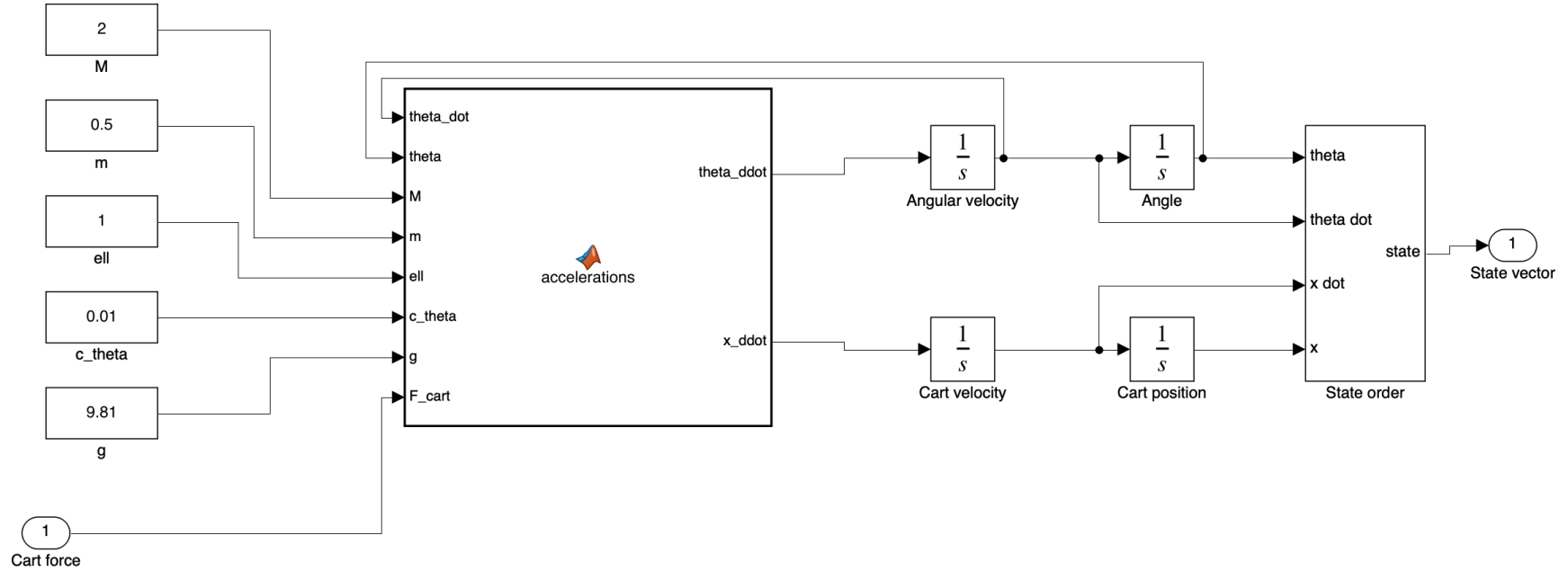


Figure 12: Nonlinear equations and four integrators retained from Project 1.

Inside the preceding feedback loop, the acceleration function from Appendix B drives two integrator chains. The State order block receives $(\theta, \dot{\theta}, \dot{x}, x)$ from top to bottom and rearranges them internally into $(x, \dot{x}, \theta, \dot{\theta})$ for the gain. The display order therefore follows the diagram, while the vector sent to the controller follows the design matrices.

References

- [1] MathWorks, “ctrb. Controllability of state-space model.” Accessed: Oct. 06, 2026. [Online]. Available: <https://www.mathworks.com/help/control/ref/statespacemodel.ctrb.html>
- [2] MathWorks, “A PID Tuning Guide. Understanding PID Control, Part 4.” Accessed: Oct. 06, 2026. [Online]. Available: <https://www.youtube.com/watch?v=sFOEsA0Irjs>
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